GREENLEAF / QUALITY ASSURANCE

CAPSTONE • OCT 2026

END-TO-END RELEASE PORTFOLIO

GreenLeaf Grocers / Checkout Overhaul

A traceable QA evidence package covering requirements, test design, defect investigation, browser automation, SQLite validation and release sign-off.

RELEASE RECOMMENDATION • NO-GO

Two documented critical defects remain open.

01 / 01	01	02	07
PLAYWRIGHT PASS	SQL ORDER VERIFIED	CRITICAL DEFECTS	EVIDENCE ITEMS

WHAT THIS DOCUMENT DEMONSTRATES

One passing guest-checkout happy path and a matching database record do not override unresolved critical release risks. The final Test Summary Report explains the evidence and the decision.

01 / TEST STRATEGY

Requirements to test evidence

RELEASE BACKLOG

GL-E01 GreenLeaf Checkout Overhaul

Scope: the guest checkout journey, promo code behavior and form validation.

TRACEABLE USER STORIES

GL-101

Guest checkout

Submit without creating an account

GL-102

Promo validation

Apply or reject promotional codes

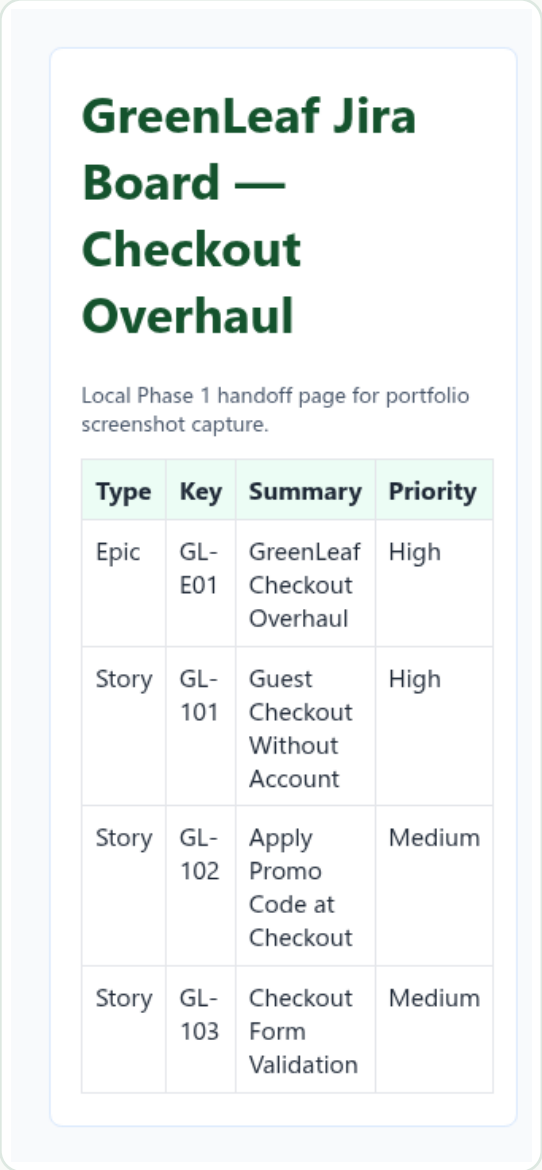
GL-103

Form validation

Prevent incomplete submissions

Coverage note. This portfolio includes one automated GL-101 happy-path run plus documented critical GL-102 defects. It is not evidence of complete release-wide test coverage.

REQUIRED EVIDENCE 01 / 07



GreenLeaf Jira Board — Checkout Overhaul

Local Phase 1 handoff page for portfolio screenshot capture.

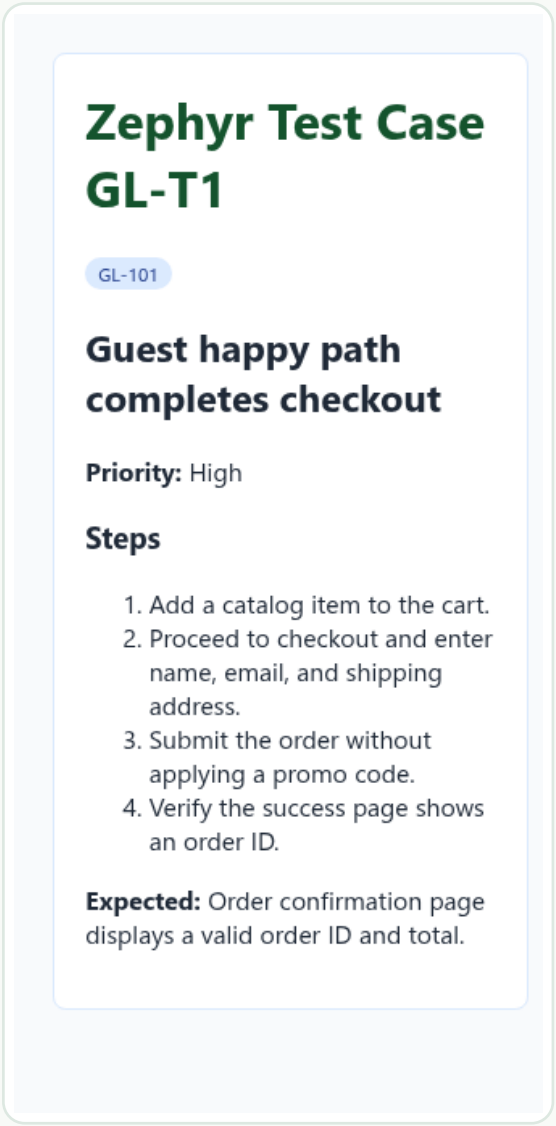
Type	Key	Summary	Priority
Epic	GL-E01	GreenLeaf Checkout Overhaul	High
Story	GL-101	Guest Checkout Without Account	High
Story	GL-102	Apply Promo Code at Checkout	Medium
Story	GL-103	Checkout Form Validation	Medium

Source: **jira_board.png** — locally rendered Phase 1 handoff board.

02 / TEST DESIGN

Guest checkout test case

REQUIRED EVIDENCE 02 / 07



Source: **zephyr_testcase.png** — GL-T1 from the supplied Zephyr-style handoff.

GL-T1 / HIGH PRIORITY

LINKED REQUIREMENT

GL-101 Guest Checkout Without Account

PRECONDITIONS

Storefront available; an item can be added to the cart; customer proceeds as a guest.

EXECUTION STEPS

1. Add a catalog item to the cart.
2. Continue to checkout and enter valid name, email and shipping address.
3. Submit the order without entering a promo code.
4. Verify the confirmation displays a non-empty order ID.

EXPECTED OUTCOME

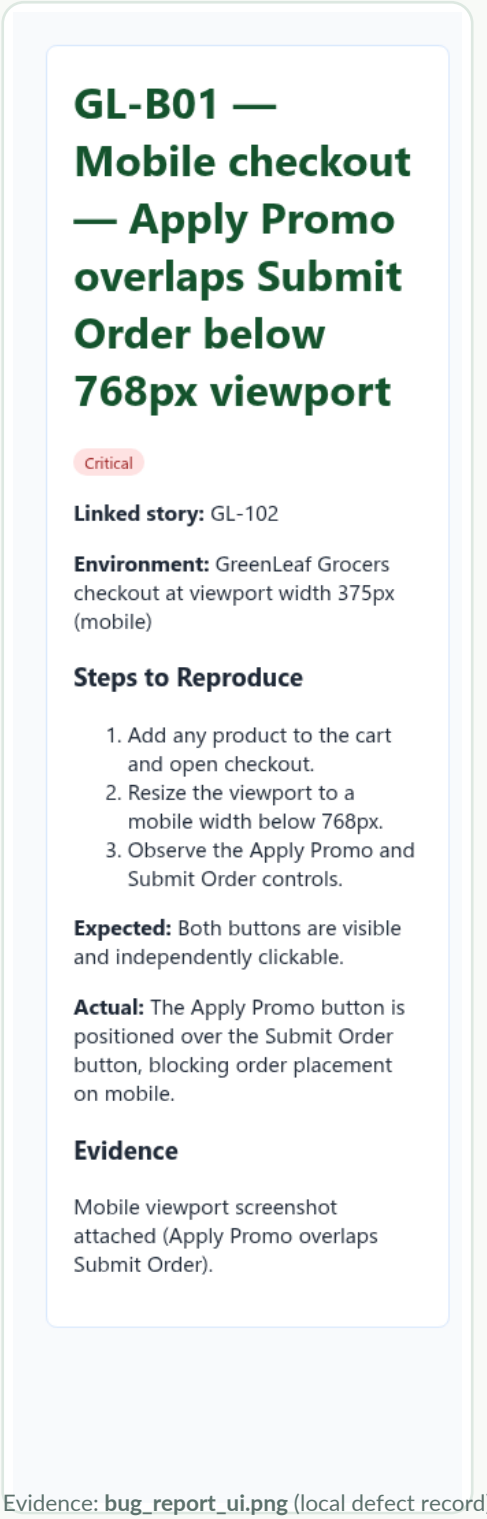
Order confirmation appears with a valid order ID and total; the checkout proceeds without account creation.

03 / DEFECT REPORT 1

GL-B01 / Mobile UI overlap

REQUIRED EVIDENCE 03 / 07

CRITICAL / OPEN



GL-B01 — Mobile checkout — Apply Promo overlaps Submit Order below 768px viewport

Critical

Linked story: GL-102

Environment: GreenLeaf Grocers checkout at viewport width 375px (mobile)

Steps to Reproduce

1. Add any product to the cart and open checkout.
2. Resize the viewport to a mobile width below 768px.
3. Observe the Apply Promo and Submit Order controls.

Expected: Both buttons are visible and independently clickable.

Actual: The Apply Promo button is positioned over the Submit Order button, blocking order placement on mobile.

Evidence

Mobile viewport screenshot attached (Apply Promo overlaps Submit Order).

Evidence: **bug_report_ui.png** (local defect record). The handoff describes the mobile overlap; an independent BrowserStack screenshot was **not** captured.

STORY & TEST ENVIRONMENT

GL-102 / GreenLeaf Grocers checkout (`/checkout`). Mobile viewport: **375 px**; affected widths below **768 px**.

DEVICE / EVIDENCE SOURCE

BrowserStack-equivalent mobile check: Chrome DevTools device emulation on the local Coursera VM. This is **not** a BrowserStack cloud capture.

ORDERED STEPS TO REPRODUCE

1. Open GreenLeaf Grocers, add one product, and navigate to Cart.

2. Select Checkout and confirm the `/checkout` page.

3. Set the viewport to 375 px width using mobile device emulation.

4. Scroll to promo and order-summary controls; inspect Apply Promo and Submit Order.

EXPECTED RESULT

Apply Promo and Submit Order remain visible, separate and clickable; checkout can proceed.

DOCUMENTED ACTUAL RESULT

The Apply Promo control overlaps Submit Order at mobile widths, obstructing checkout.

USER IMPACT

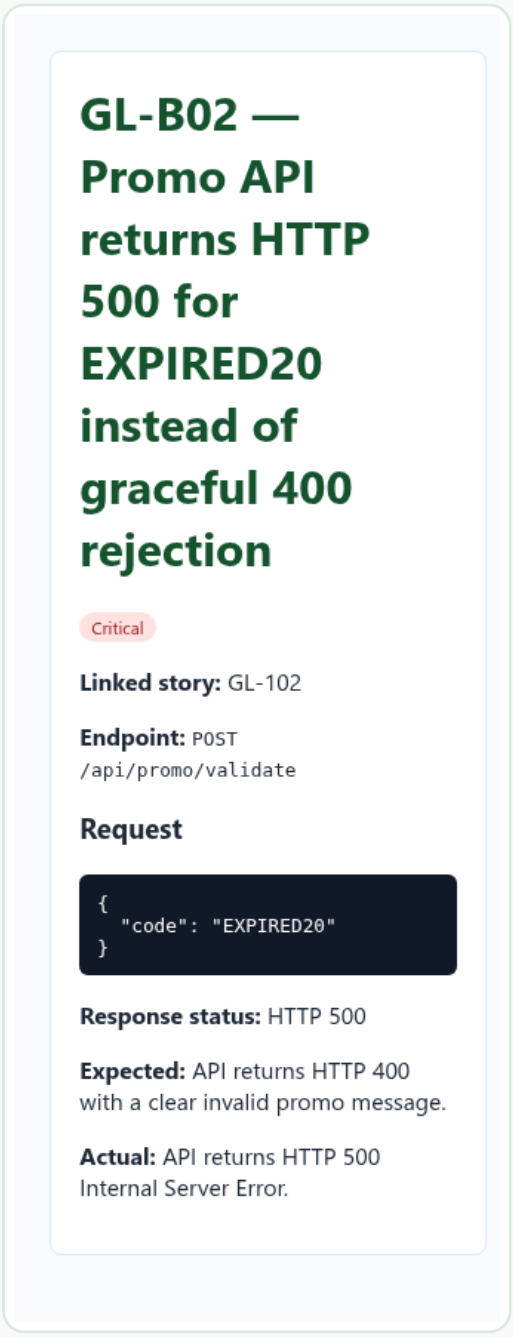
Mobile shoppers may be unable to submit their orders. Critical release-blocking risk.

Verification limitation: documented in the supplied GL-B01 handoff; not independently reproduced during this session.

04 / DEFECT REPORT 2

GL-B02 / Expired promo returns 500

REQUIRED EVIDENCE 04 / 07



Evidence: **bug_report_api.png** (Phase 1 local ticket). Direct HTTP probe also returned **500** for the expired code.

CRITICAL / OPEN

STORY & API IDENTIFICATION

GL-102 / promo codes. The grading rubric calls this area **/api/promos**; the actual endpoint supplied and tested by GreenLeaf is **POST /api/promo/validate**.

PRECONDITIONS

App running at localhost:3000; expired promo code **EXPIRED20** available for validation.

STEPS TO REPRODUCE

1. Send POST to `/api/promo/validate`.

2. Set Content-Type: application/json and body `{"code":"EXPIRED20"}`.

3. Inspect HTTP status and response JSON.

EXPECTED RESULT

HTTP 400 Bad Request (or another controlled 4xx) with a clear "promo code expired" message; no server error.

ACTUAL RESULT

HTTP 500 Internal Server Error; JSON error code `internal_error`, message "Internal server error."

CUSTOMER IMPACT

Expired codes produce a server failure rather than clear guidance; fix and retest before release.

Accuracy note: the local VM did not expose a tested `/api/promos` route; the observed request used `/api/promo/validate`.

05 / AUTOMATION

Playwright / Guest happy path

1 TEST PASSED

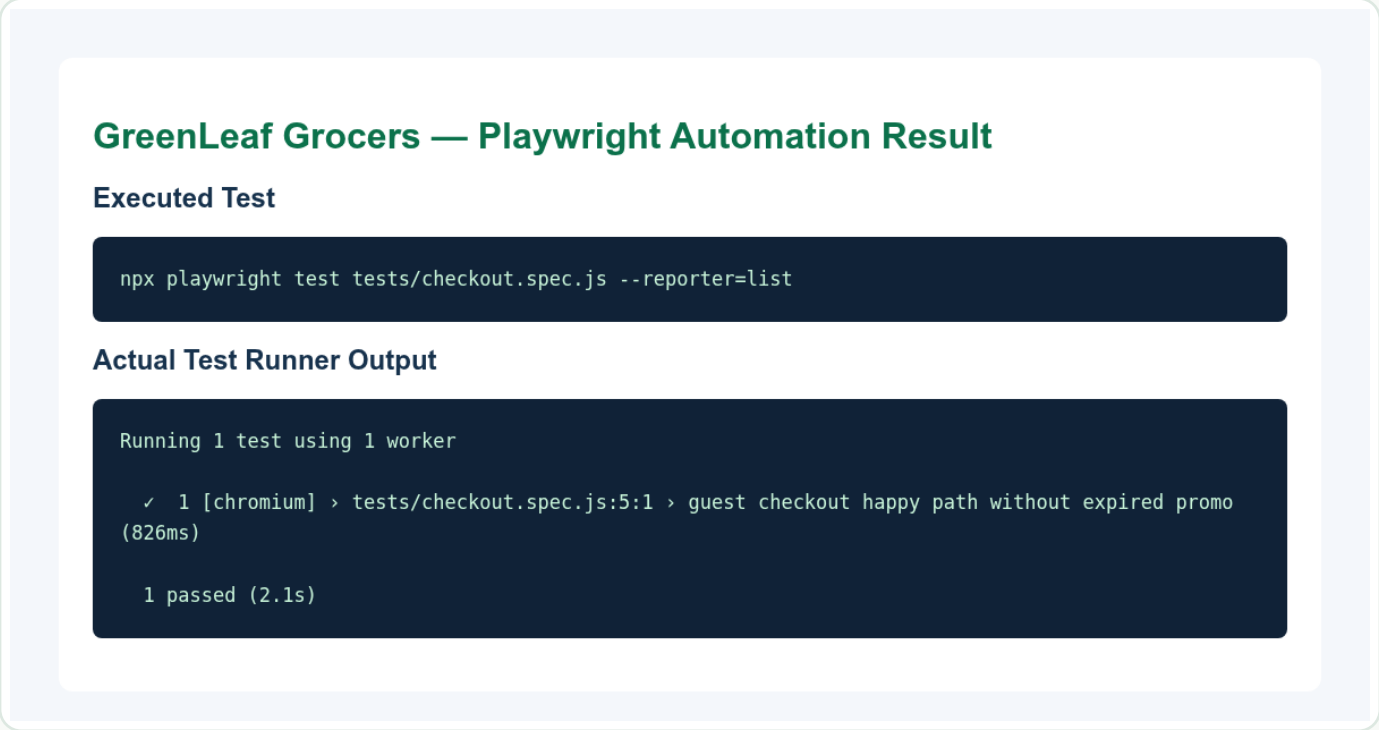
Chromium • Single happy-path execution

EXECUTED FLOW

Catalog → Add product → **Cart** → Checkout → Enter guest details → Submit order → Confirm non-empty order ID.

Test selector strategy: stable `data-testid` attributes. The expired promo code was intentionally excluded.

REQUIRED EVIDENCE 05 / 07



Source: **playwright_pass.png**. This image was generated from a real Playwright test-run log; it is a formatted evidence view, not a direct VM terminal capture.

WHAT THIS PROVES

The chosen guest-checkout route completed successfully in one executed Chromium run. It does **not** demonstrate the mobile promo defect or expired-promo API behavior is fixed.

06 / BACKEND VALIDATION

SQL / Order persistence

QUERY LOGIC

JOIN `orders` to `users` on `orders.user_id = users.id`; filter the guest email, sort newest first, return one row.

REQUIRED EVIDENCE 06 / 07



Source: **sql_validation.png**. This composed screenshot displays the executed SQL text and its actual saved SQLite result.

VALIDATED FIELDS

EMAIL	ORDER STATUS
capstone.guest@example.test	placed

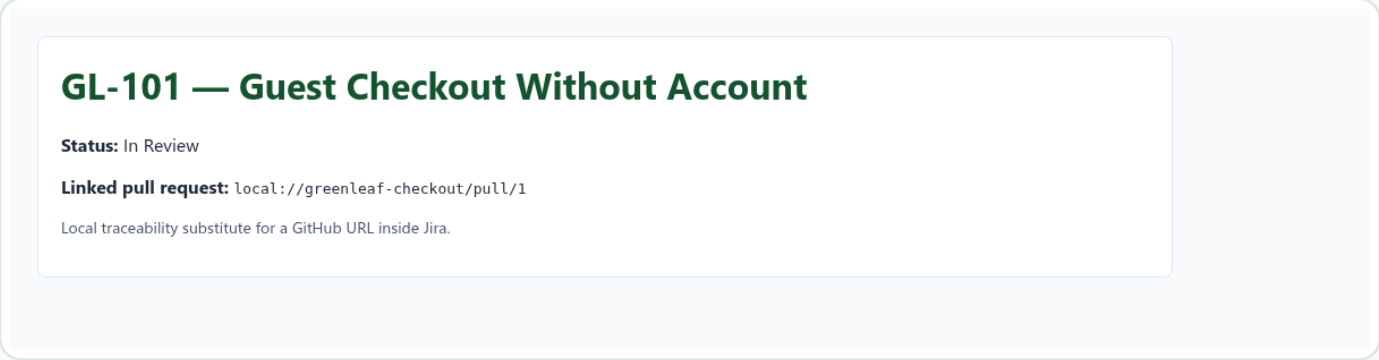
TOTAL 4.99

Result: a matching newest order row exists. This closes the UI-to-database validation loop for the tested happy path.

07 / VERSION CONTROL & JIRA PR

Traceability / Local pull request

REQUIRED EVIDENCE 07 / 07



Source: **jira_pr_link.png** — local Jira-style story view. `local://` is a simulator reference, not a public GitHub URL.

TRACEABILITY CHAIN

GL-101	GL-T1	Playwright	Local PR #1	In Review
Requirement	Test case	Automation	Code review	Jira status

VERIFIED RECORD

Story GL-101 — Guest Checkout Without Account

Status In Review

Branch feature/checkout-happy-path

Linked request local://greenleaf-checkout/pull/1

The Coursera VM uses a nested local Git repository and PR/ticket records. This is evidence of a simulated Git/Jira workflow rather than a hosted GitHub pull request.

FINAL SECTION / AI-ASSISTED REPORT

Test Summary Report

GreenLeaf Checkout Overhaul • Stakeholder release-readiness assessment

RELEASE RECOMMENDATION

NO-GO

Hold release until both documented critical defects have been corrected, retested and assessed for regression impact.

EXECUTIVE SUMMARY

The GreenLeaf Grocers guest checkout happy path completed successfully in Playwright, and a targeted SQLite query returned a persisted order record for **capstone.guest@example.test**. However, the course-provided handoff identifies two unresolved critical checkout defects. The passing automation result supports the tested path only; it does not establish overall release readiness.

SCOPE & EVIDENCE

This portfolio covers backlog traceability (GL-E01 / GL-101–103), the GL-T1 guest checkout test case, two critical defect records, one executed Chromium happy-path test, one order-validation SQL query, and local Git/PR-to-Jira traceability. The report is based on the seven included evidence images and verified lab results.

TEST EXECUTION & VALIDATION

Playwright guest checkout	**1 executed / 1 passed**
SQLite persistence	**Newest matching order found**
Guest order status	**placed, total 4.99**
Open critical defects	**2 documented in handoff**

No broader regression suite, repaired-defect retest or full device-matrix result is evidenced by this package.

TEST SUMMARY REPORT / CONTINUED

Risks, limitations & sign-off

OPEN CRITICAL DEFECTS

GL-B01 / CRITICAL	**Mobile checkout layout:** the Phase 1 handoff reports Apply Promo overlapping Submit Order below 768 px, potentially blocking mobile checkout. The training handoff provides this finding. BrowserStack-equivalent DevTools emulation was specified; an independent cloud BrowserStack screenshot is not included.
GL-B02 / CRITICAL	**Expired-promo API failure:** `POST /api/promo/validate` with `EXPIRED20` returned HTTP 500 instead of a controlled 4xx validation response. This was observed at the actual GreenLeaf lab route (rubric category /api/promos) in the Phase 1 API probe.

BUSINESS IMPACT & TEST LIMITATIONS

The mobile issue threatens order completion, while the API failure creates an avoidable server error during promotion use. The passing non-promo guest journey and matching SQLite record are positive results, but neither proves mobile layout or expired-code recovery. Defect fix verification, additional checkout variants and cross-device regression remain outside the evidence provided.

REQUIRED ACTIONS BEFORE GO

- 1. Repair GL-B01 and verify accessible, non-overlapping mobile checkout controls at and below the affected breakpoint.
- 2. Repair GL-B02 so expired promo validation returns a controlled client error and an understandable customer-facing message.
- 3. Retest both defects, rerun the guest checkout automation, and repeat order-persistence SQL validation.
- 4. Record updated evidence and request a fresh release-readiness review.

FINAL QA SIGN-OFF

NO-GO recommended. The release should remain on hold until GL-B01 and GL-B02 are resolved and successful retest evidence is reviewed. Final deployment authorization rests with the release stakeholders.

Methodology note: this is a course lab using local Jira/Zephyr/PR equivalents. The Test Summary Report was produced from structured, AI-assisted lab reporting and then editorially refined for accuracy and clarity.